

Jiaju Wu

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Education

• Sun Yat-sen University

M.S. in Physics (Optics), School of Science (Recommended Admission)

Sep 2025 – Present

- Grade: 91/100, supervisor: Assoc. Prof. Xiantao Jiang

B.S. in Physics, School of Physics and Astronomy

Sep 2021 – Jun 2025

- GPA: 3.6/4.0, Rank: 28/113

- Outstanding Bachelor's Thesis (Top 7%)

Research Experience

Research areas: Structured-light speckle, OAM photonics, Deep learning, scattering imaging and quantitative phase microscopy

• SYSU | Optics Imaging Lab

Feb 2025 – Present

• Project 1: *Dual-Dimensional Recognition of OAM and Geometry of Generalized Perfect Optical Vortices in Strong Scattering via Correlation-Guided Deep Learning*

- Revealed unique correlation features of Generalized Perfect Optical Vortex (GPOV) speckles in strong scattering environments
- Developed Correlation-Guided Residual Network (CG-ResNet) for accurate and fast simultaneous OAM mode and geometry recognition from GPOV speckles, achieving >93.33% OAM accuracy and >98.75% geometry accuracy
- Achieved superior generalization across diverse scattering conditions with low computational cost

• Project 2: *High-Order Orbital Angular Momentum Extraction from Vortex-Beam Speckles with Low Spatial Sampling*

- Revealed the physical mechanism behind speckle correlation degradation for high-order OAM modes
- Proposed a low-complexity topological charge-matching strategy to extend the accessible range of high-order OAM retrieval
- Realized accurate OAM extraction with only 0.3% speckle pattern maintaining robustness against strong scattering

• Project 3: *Simulation and Measurement of the Point Spread Function of Laser Speckle Patterns* (B.S. Thesis, Outstanding)

- Conduct numerical simulations on the formation and propagation of speckles, and experimentally measure the speckle point spread function and the range of optical memory effect
- Realized speckle autocorrelation imaging based on phase retrieval algorithms (GS, HIO-ER); achieved speckle PSF deconvolution imaging by adopting Lucy-Richardson and Wiener filtering algorithms

Supervisor: Assoc. Prof. Xiantao Jiang

• SYSU | MOE Key Laboratory of TianQin Mission

Nov 2023 – Oct 2024

• Project 4: *Theoretical Modeling and Numerical Simulation of Multi-Beam Interference Based on the Astigmatic Gaussian Beam Model* (Undergraduate Innovation Training Program, Project Leader)

- Based on the astigmatic Gaussian beam model, numerical simulations of multi-beam interference and stray light noise evaluation for the TianQin Project laser interferometer are carried out
- Quantitative calculation of the suppression effect of the balanced detection scheme on stray light noise in a picometer sensitivity dual-frequency heterodyne laser interferometer

Supervisor: Assoc. Prof. Huizong Duan

Publications

The first-author:

- **Jiaju Wu**, Rui Cao, Shengqiang Zhong, Yuhan Liu, Kaibin Zeng, Rushi Yang, Li Yang, Xiantao Jiang. "High-Order Orbital Angular Momentum Extraction from Vortex-Beam Speckles with Low Spatial Sampling," *Under review at Optics & Laser Technology*, 2026.
- **Jiaju Wu**, Kaibin Zeng, Shengqiang Zhong, Yuhan Liu, Rushi Yang, Xiantao Jiang. "Dual-Dimensional Recognition of OAM and Geometry of Generalized Perfect Optical Vortices in Strong Scattering via Correlation-Guided Deep Learning," *Manuscript in Preparation*, 2026.

Other:

- Kaibin Zeng, Weiyuan Li, Wenjun Zhang, Bin Liu, **Jiaju Wu**, Shengqiang Zhong, Hongwei Zou, Fan Yang, Chao Hou, Lei Xu, Xianglong Yu, Xin Luo, Xiantao Jiang. "Precise characterization of optical constant of two dimensional WS2 using diffraction phase microscopy," *Optics & Laser Technology* 192, 114109 (2025). (JCR Q1, IF=5)
- Yuhan Liu, Hongwei Zou, Shengqiang Zhong, **Jiaju Wu**, Kaibin Zeng, Rushi Yang, Xiantao Jiang. "Structure light-modulated speckle field for three-dimensional high-resolution biomedical imaging," *Submitted for publication*, 2026.

Honors & Awards

• First-class Scholarship of Sun Yat-sen University	2025
• Second-class Scholarship of Sun Yat-sen University	2024
• Meritorious Winner (Top 7%) in The Interdisciplinary Contest in Modeling (ICM)	2024
• Project Leader of Undergraduate Innovation Training Program Grant (Funded by SYSU)	2023
• Third-class Scholarship of Sun Yat-sen University	2023
• Secondary Prize of The Mathematics competition of Chinese College Students	2022
• Secondary Prize of The Asia and Pacific Mathematical Contest in Modeling	2022
• Special Scholarship of Sun Yat-sen University (Outstanding Conduct Award)	2022
• First Prize (Team Leader) of China Undergraduate Physics Tournament (CUPT, SYSU Selection)	2021

Skills

Languages: English (CET-6: 566), Mandarin Chinese (native)
Programming and Tools: MATLAB, Python (PyTorch, scikit-learn...), C++, Lumerical FDTD, COMSOL, Origin, LaTeX, Blender

Teaching / Mentorship

Teaching Assistant: *Probability and Statistics for Science and Engineering* (Autumn 2025)
Mentorship: Mentored an undergraduate research project on *Speckle Formation Mechanisms Using FDTD Simulations*.